$$\therefore \int_a^b f(x)dx = \int_a^b f(a+b-x)\,dx$$

Evaluate the following using properties of definite integrals:

Example 2.75

Evaluate $\int_{-1}^{1} \frac{x^5\,dx}{a^2-x^2}$

Solution:

Let $f(x) = \frac{x^5}{a^2-x^2}$

$$f(-x) = \frac{(-x)^5}{a^2-(-x)^2}$$

$$= \frac{-x^5}{a^2-x^2}$$

$$= -f(x)$$

Here $f(-x) = -f(x)$

$\therefore f(x)$ is an odd function

$$\Rightarrow \int_{-1}^{1} \frac{x^5\,dx}{a^2-x^2} = 0$$

Example 2.76

Evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x\,dx$

Solution:

Let $f(x) = \cos x$

$f(-x) = \cos(-x) = \cos x$

$\Rightarrow f(-x) = f(x)$

$\therefore f(x)$ is an even function

$$\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x\,dx = 2\int_0^{\frac{\pi}{2}} \cos x\,dx$$

$$= 2\left[\sin x\right]_0^{\frac{\pi}{2}}$$

$$= 2\left[\sin\frac{\pi}{2} - \sin 0\right]$$

$$= 2$$

Example 2.77

Evaluate $\int_{-1}^{1} \left(x^2+x\right)dx$

Solution:

$$\int_{-1}^{1} \left(x^2+x\right)dx = \int_{-1}^{1} x^2\,dx + \int_{-1}^{1} x\,dx$$

$$= 2\int_0^1 x^2\,dx + 0 \quad [\because x^2 \text{ is an even function and } x \text{ is an odd function}]$$

$$= 2\left[\frac{x^3}{3}\right]_0^1 = 2\left[\frac{1}{3}-0\right]$$

$$= \frac{2}{3}$$

Example 2.78

Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x+\cos x}dx$

Solution:

Let $I = \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x+\cos x}dx \quad \ldots(1)$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin\left(\frac{\pi}{2}-x\right)}{\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{\pi}{2}-x\right)}dx$$

$$\left[\because \int_0^a f(x)dx = \int_0^a f(a-x)dx\right]$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos x}{\cos x+\sin x}dx \quad \ldots(2)$$

$(1)+(2) \Rightarrow$

$$2I = \int_0^{\frac{\pi}{2}} \left[\frac{\sin x}{\sin x+\cos x} + \frac{\cos x}{\cos x+\sin x}\right]dx$$

$$= \int_0^{\frac{\pi}{2}} dx = \left[x\right]_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

$\therefore \quad I = \frac{\pi}{4}$

XII Std - Business Maths & Stat EM Chapter 2.indd 49 2/28/2020 5:40:58 PM